Student Enrollment System

CS 211 — File Processing

Team Members

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Project Description

The Student Enrollment System is a file-based management system designed to store, search, delete, and display student and course information using text files.

The system aims to provide a simple and effective interface that replaces manual record-keeping.

Objectives

- **Add new student and course records**
- **Search for existing records**
- **Display all records**
- **Delete records from file**
- **Use text files to store data permanently**
- **Provide user-friendly Windows Forms interface**

Usefulness

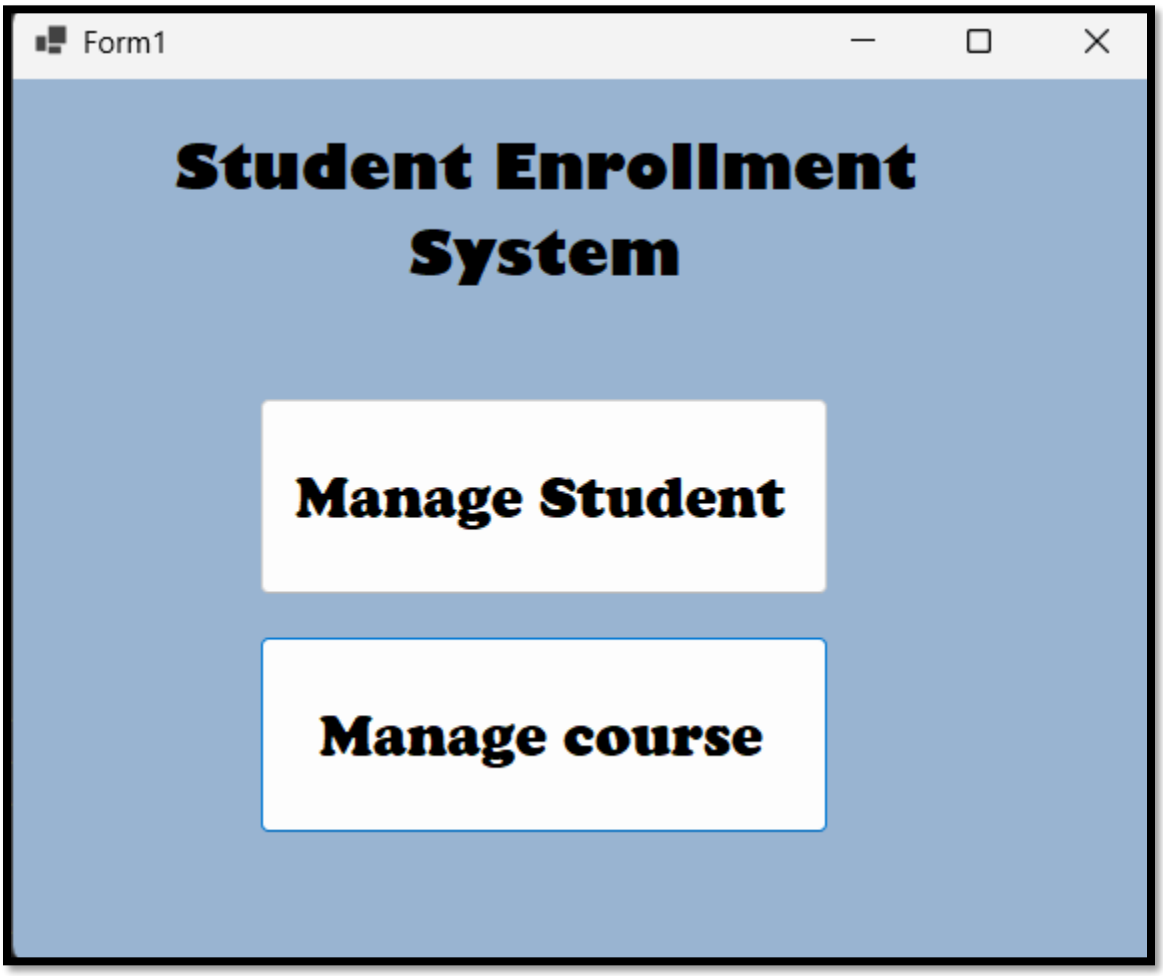
The system helps instructors and administrators store and manage data in organized files instead of using handwritten notes or Excel.

Problems Solved

- **Eliminates manual errors**
- **Saves time**
- **Stores data safely**
- **Easy to update and search**

System Scenarios

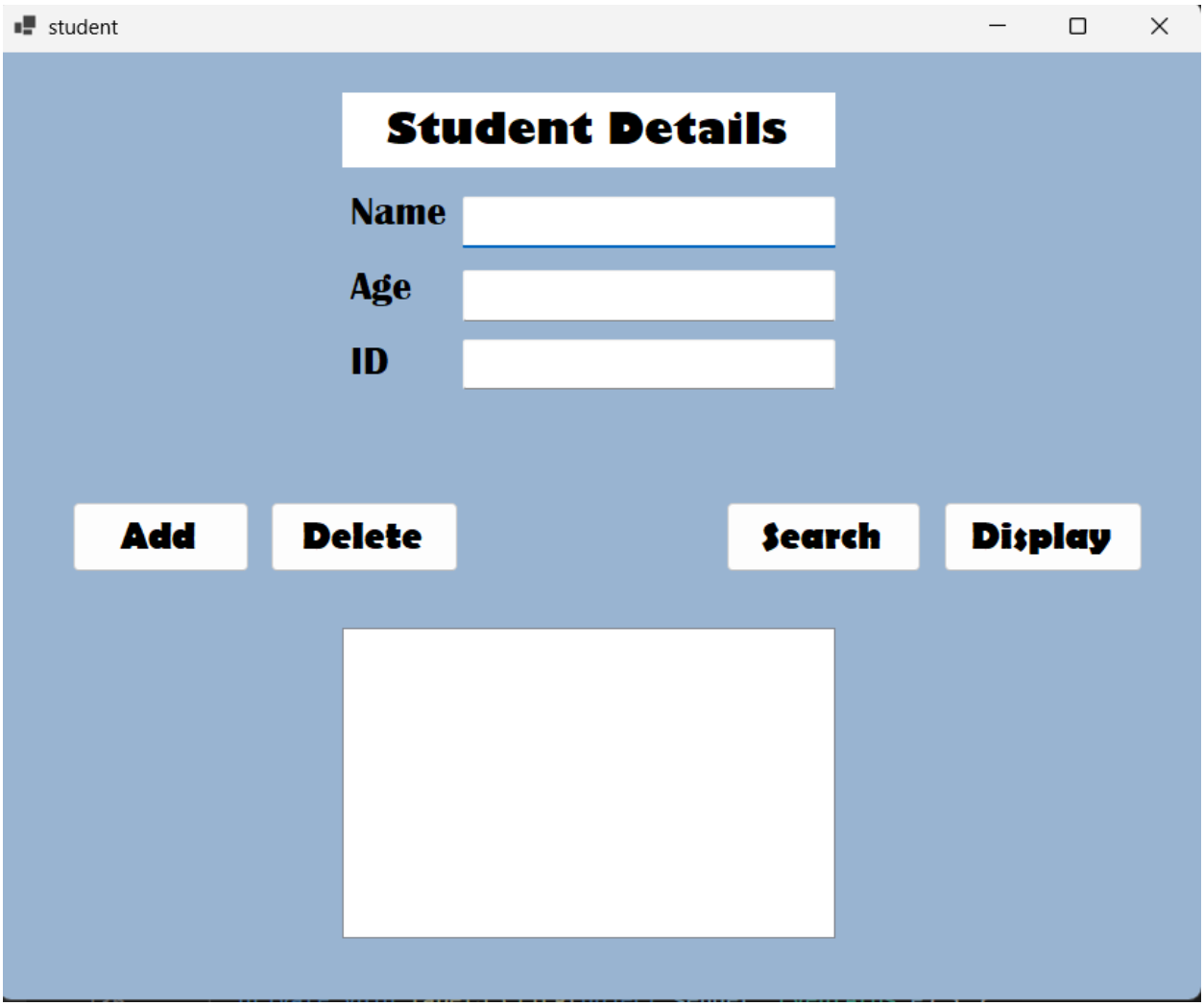
1. Main Menu



User opens the program → Main menu appears with two choices:

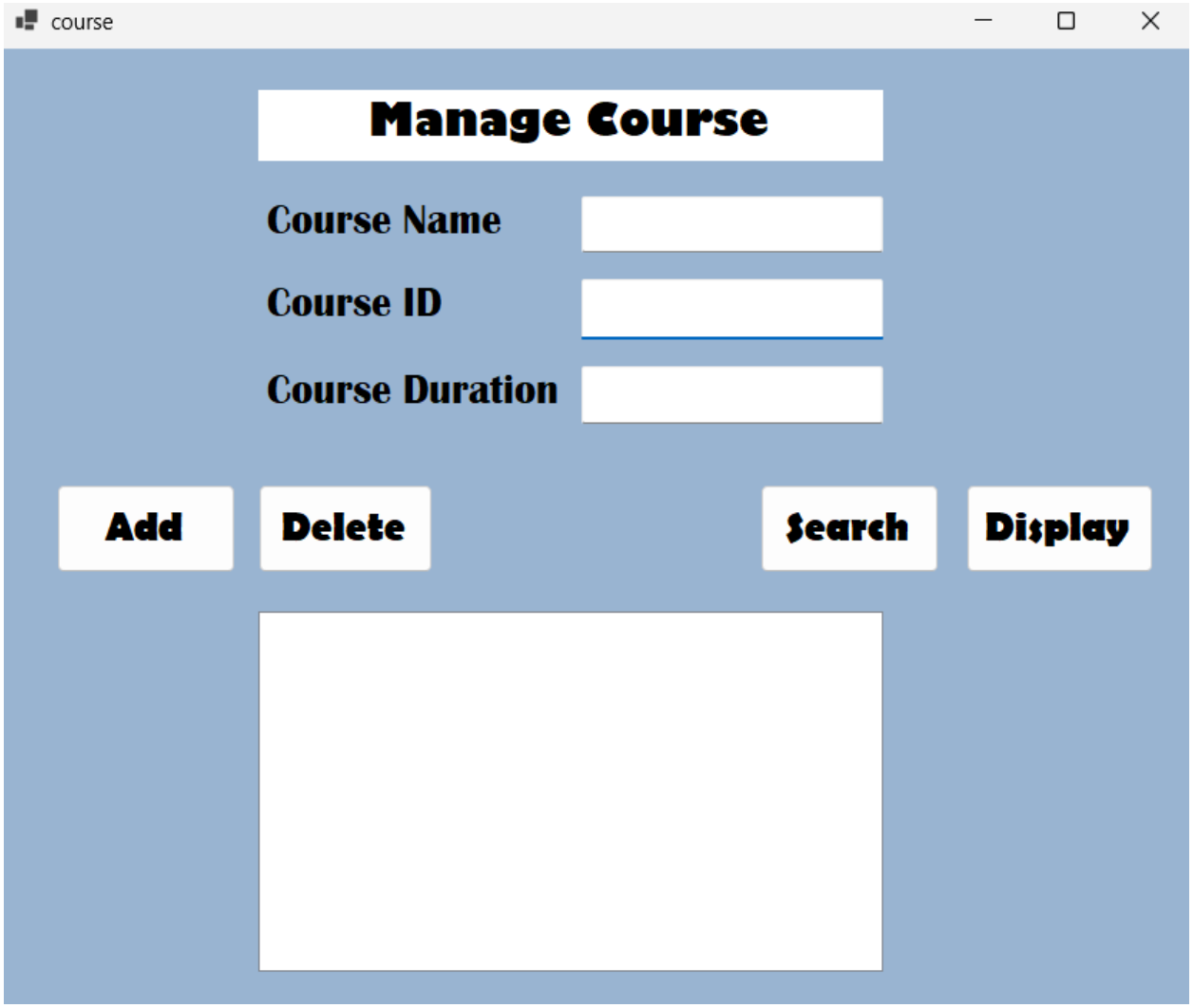
1. Manage Students
2. Manage Courses

2. Student Form



1. Add Student
2. Delete Student
3. Search Student
4. Display Student

1. Course Form



1 - Add Course

2 - Delete Course

3 - Search Course

4 - Display Course

Code Snippets

```
70
71          // Search
72          private void button3_Click(object sender, EventArgs e)
73          {
74              string courseId = textBox1.Text.Trim();
75
76              if (!File.Exists(path))
77              {
78                  MessageBox.Show("File not found!");
79                  return;
80              }
81
82              string[] lines = File.ReadAllLines(path);
83              bool found = false;
84
85              listBox1.Items.Clear();
86
87              foreach (string line in lines)
88              {
89                  string[] data = line.Split(',');
90
91                  if (data[0].Trim() == courseId)
92                  {
93                      listBox1.Items.Add($"ID: {data[0].Trim()} | Name: {data[1].Trim()} | Duration: {data[2].Trim()}");
94                      found = true;
95                      break;
96                  }
97              }
98
99              if (!found)
100                 MessageBox.Show("Course not found!");
101         }
```

```
37
38          // Delete
39          private void button2_Click(object sender, EventArgs e)
40          {
41              string courseId = textBox1.Text.Trim();
42
43              if (!File.Exists(path))
44              {
45                  MessageBox.Show("File not found!");
46                  return;
47              }
48
49              string[] lines = File.ReadAllLines(path);
50              List<string> newList = new List<string>();
51              bool deleted = false;
52
53              foreach (string line in lines)
54              {
55                  string[] data = line.Split(',');
56
57                  if (data[0].Trim() != courseId)
58                      newList.Add(line);
59                  else
60                      deleted = true;
61              }
62
63              File.WriteAllLines(path, newList);
64
65              if (deleted)
66                  MessageBox.Show("Course Deleted!");
67              else
68                  MessageBox.Show("Course ID Not Found!");
69          }
```

```
18
19          // Add
20          private void button1_Click(object sender, EventArgs e)
21          {
22              string courseId = textBox1.Text.Trim();
23              string courseName = textBox3.Text.Trim();
24              string duration = textBox2.Text.Trim();
25
26              if (courseId == "" || courseName == "" || duration == "")
27              {
28                  MessageBox.Show("Please fill all fields!");
29                  return;
30              }
31
32              string line = $"{courseId},{courseName},{duration}";
33              File.AppendAllText(path, line + Environment.NewLine);
34
35              MessageBox.Show("Course Added Successfully!");
36          }
37
```

```
// Display
private void button4_Click(object sender, EventArgs e)
{
    listBox1.Items.Clear();

    if (!File.Exists(path))
    {
        MessageBox.Show("File not found!");
        return;
    }

    string[] lines = File.ReadAllLines(path);

    foreach (string line in lines)
    {
        string[] data = line.Split(',');
        if (data.Length >= 3)
        {
            listBox1.Items.Add($"ID: {data[0].Trim()} | Name: {data[1].Trim()} | Age: {data[2].Trim()}");
        }

    }
}
```

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